

Chapter – 11: Nature's Treasures

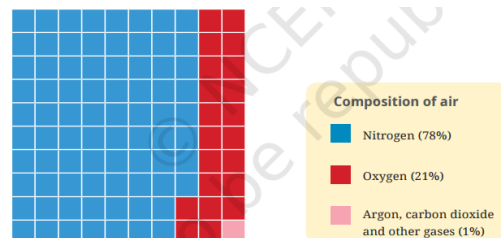
- School vacations – always fun – Bhoomi, Surya – visiting *Aji* (grandmother)
- *Aji* – village – edge of forest – Western Ghats
- Village air – fresh, cooler – beautiful hills, streams, animals, plants, etc.
- One afternoon – Bhoomi, Surya – ask *Aji* – tell about place
- *Aji* – tells children – natural treasures – enrich our lives – pure air – refreshing – fertile soil – supports variety of living beings
- This place – plenty of sunlight – useful – many ways – trees – provide food, shelter – animals, insects
- Bhoomi – we need water – drinking, growing vegetables
- *Aji* – these treasures – req. – survival, making lives more comfortable
- Without these – any form – life – not possible

Air

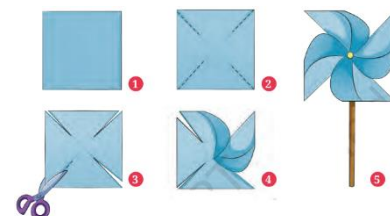
- One morning – Bhoomi, Surya – obs. *Aji* – breathing exercises
- *Aji* explains – deep breath in, breath out – helps – more fresh air – inside lungs – stay healthy
- Activity –

- Take deep breath in, breath out
- Deep breath in again – hold your breath – as long you can – breath out slowly
- Obs. – how long – can hold breath
- Also obs. – feelings – holding breath

- This act. – find out – difficult – hold breath for long
- Air – breath in – oxygen – body need it – perform func.
- Hold breath – longer period – no oxygen for body – no func. – have to release breath
- Air – surrounding earth – mixture of gases – nitrogen, oxygen, argon, carbon dioxide, other gases
- Fig. – 100 sq. – 78 sq. – nitrogen – 21 sq. – oxygen – 1 sq. – argon, carbon dioxide, other gases
- Notice – air presence – tree leaves – rustle
- Moving air – wind – sometimes fast (storm) – sometimes slow (breeze)



- Activity –
- Take sq. paper – 15 x 15 cm – pair of scissors – all-pin – soft stick
- Follow instructions – given fig. – make a *firki*



- Now – hold *firki* – run – can also blow air
- Move a little – back and forth – *firki* rotates – due to wind
- Working – windmill – similar – wings – rotated by wind
- Windmill – used – run flour mills, pull up water, generate electricity
- India – many windmill farms – area – large no. of windmills – generate electricity

Water

- Bhoomi, Surya – help *Aji* – filling troughs – water for cows
- Also help her – watering plants – vegetables, medicinal herbs

- *Aji* – teaches them – how to water – utilise every drop
- Need water – many daily act. – drinking, cooking, bathing, washing, cleaning
- Also used – growing crops, industrial purposes
- Water – covers – $\frac{2}{3}^{\text{rd}}$ – earth's surface – most of it – oceans, seas
- This water – saline, salty – not fit – domestic, agricultural, industrial uses
- All these act. – req. freshwater – ice sheets, snow, rivers, lakes, underground water, etc.
- These src. – difficult to access – very small fraction – freshwater – available
- Water – very precious – reason – *Aji* – asks kids – use it wisely
- Many ways – reduce wastage – water
- Ex. – turn off taps – not in use – fixing water leakages – recycling water – water harvesting
- Our country – blessed – numerous rivers, streams, lakes
- People – pollute freshwater – throwing trash
- Waste – homes, industries – pollute water src.
- Polluted water – not fit – consumption
- Freshwater src. – limited – shortage of water – many parts
- Some places – people – walk long dist. – fetch drinking water
- Everyone – not same access – imp. – conserve water – use it judiciously
- Also protect – from pollution – remains available – everyone
- Water harvesting – method – conserving water
- Many buildings – rainwater – collected, stored – large qty. – **rainwater harvesting**
- Stepwells – *Bawadi* – Rajasthan – *Vav* – Gujarat – built – water harvesting – response – scarcity of water
- Stepwells – unique sys. – store rainwater – also seeping water from nearby lakes, ponds, rivers
- Walls – trenches – lined with blocks of stones – allow seepage

Energy from the Sun

- Sunny day – Bhoomi, Surya – helping *Aji* – dry chillies
- *Aji* explains – sun heat – dries fresh chilly – energy from sun – main src. – earth – everyone req. it
- Sun – heat, light – various purposes
- Afternoon – Bhoomi, Surya – pass by field – cow grazing
- They talk – sun – main src. – energy
- Bhoomi – cow grazing – getting energy – grass
- Surya – no, cow getting energy – sun
- Bhoomi – yes, cow – standing in sun – BUT – does not mean – getting energy – sun
- Surya – cow – eating grass – grass – need sunlight – this way – cow – getting energy – sun – main src. – energy
- Sunlight – helps plants – prepare food – also helps – living beings – heat, light
- Solar panels – capture energy – sun – generate elec.
- Sun energy – used directly – cook food (solar cookers) – heat water (solar water heaters)
- No life on earth – without sunlight – main src. – energy
- Plants – get energy – sun – produce food
- Animals – eat plants – grow
- Humans – get food – both – plants, animals
- All dependent – sun – main src. – energy

Forests

- Morning – *Aji* takes – Bhoomi, Surya – walk in forest
- They find – variety – herbs, shrubs, trees
- *Aji* explains – forests – large areas – dense growth – various plants
- On the way – they collect – *nellikai* (Indian gooseberries) – fallen onto ground
- *Aji* explains – village tradition – do not pluck fruits from trees – they are for animals, birds
- Forests – natural home – wild animals, birds, insects – provides shelter, food
- Nature – every animal – depends on other life forms – survival
- Diversity of life forms – ensure food – everyone
- Over yrs. – forest cover – decreasing – mainly – human act. – large scale cutting of trees
- Many yrs. – grow, restore forest – must preserve – use responsibly
- *Van Mahotsav* – week long event – July month – forest festival – new plants, trees – planted
- Aim – this event – increase green cover
- During walk – Bhoomi, Surya – notice – lots of leaves – soil feels damp
- *Aji* explains – roots – hold soil – prevent from washing away
- Leaves – fall from trees – decay, enrich soil – nutrients
- New plants, trees – use this soil – grow – ex. – recycling in nature

Soil, Rocks and Minerals

- Bhoomi, Surya, *Aji* – return home
- Bhoomi, Surya – help *Aji* – prepare soil – garden – planting vegetables
- *Aji* – asks them – dig soil – gently loosen lumps
- Space b/w soil particles – provides sufficient air – also provides – space for roots
- Bhoomi, Surya – notice – small pebbles, roots, earthworms
- Activity –
 - Collect – samples – soil – diff. areas – home, school
 - Guess – what's there – diff. soils
 - Obs. – each soil sample – note – colour
 - Touch – each soil sample – feel – texture
 - Obs. – soil samples – naked eye – look through – magnifying glass
 - Record obs.
- Soil – many things – sand, insects, worms
- Many small org. – cannot see – naked eye
- Plants, animals – become part – soil – decompose, decay
- Soil samples – diff. places – diff. colours – diff. materials
- Soil – formed – disintegration (breaking apart) – rocks – actions – sun, water, living org. – long time
- Diff. soils – some – good – certain crops – others – good – making bricks
- Forest – variety of soils – precious treasure – supports biodiversity
- Obs. surroundings – rocks – used – construction – houses, buildings, temples, roads, dams, table tops
- Some rocks – slate – used – roofing – laterite – used – building material
- Imp. rocks – granite, sandstone, marble
- Humans – using rocks – making tools – hand axes, arrowheads – 1000s of yrs.
- Rocks – made of – minerals – imp. metals – aluminium, gold, copper, iron
- Minerals – used – manufacture – airplanes, jewellery, cosmetics, electrical, electronic equipment

- Ex. – basic mobile phones – dozen materials – gold, silver, copper, cobalt, etc.
- Rocks – vital role – our lives – 1000s to millions yrs. – form rocks
- Imp. – conserve, use responsibly

Fossil Fuels

- Activity –
 - Conduct survey – vehicles – neighbourhood
 - Types of vehicles – types of fuels – they use?
 - Record – info.
- Petrol, diesel – 2 most widely used fuels
- Petrol, diesel, kerosene – obtain from petroleum – along with – natural gas, coal – **fossil fuels**
- Formed – remains – microorganisms, plants – buried – deep inside earth
- Millions of yrs. – form these fuels
- Natural gas – used – cooking – generate elec.
- Now – used in form – Compressed Natural Gas (CNG) – fuel – vehicles
- CNG – much cleaner – petrol, diesel
- Coal – mainly used – production – elec. – found – several parts – India
- Fossil fuels – limited qty. – soon – run out – use uncontrollably
- Avoid this situation – explore – alt. src. – energy
- Fossil fuels – burnt – smoke, CO₂ – pollutes air
- Over-dependence – fossil fuels – transportation, domestic fuels – large scale pollution

Natural Resources – Renewable and Non-Renewable

- Nature's treasure – fulfill – needs – essential – sustain life forms
- Ex. – sun – heat, light – rivers – water – plants, animals – food
- Resources – from nature – **natural resources**
- Natural resources – used – many useful things – make life – convenient – **human-made resources**
- Natural resources – air, water, sun energy, forest, soil, rocks, minerals, fossil fuels
- Some resources – replenish – natural processes – period of time
- *Aji* – told Bhoomi, Surya – collect *nellikai* – fallen on ground – make sure – enough on trees – other animals, birds
- Seeds – droppings – animals – grow new trees
- Resources – replenished – reasonable period of time – **renewable resources** – air, water, forest
- Fossil fuels – replenish – millions of years – once used – exhausted – do not replenish – reasonable period of time – **non-renewable resources** – minerals, rocks, soil, coal, petroleum

Resources We Use

- Its time – Bhoomi, Surya – go home – after vacation – *Aji*'s home
- *Amma* (mother) – picks them up – both – show her – vegetable plants, dried chilly
- They notice – change in skyline colour – change in air – reaching city
- Very few trees – air – does not smell good – smoke can be obs.
- We must use resources – responsibly – fulfill our need – save for future